

Field Artillery Operations

INSTRUCTOR'S PROFILE



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PREVIOUS ASSIGNMENTS:

- Btry Cmdr, B Btry, 4th Field Artillery Battalion, DS 402Bde, Butuan City
- Chief, Doctrine Br, FA Capability Development Office
- Head, Training Dept. Artillery Training School
- Opns Officer, Artillery Training School
- Civil-Military Operations Officer, 71st Infantry Battalion, Comval
- Pltn Ldr, Mrtrt & Sct Pltn, 71st Infantry Battalion, ComVal

MILITARY SCHOOLING:

- Staff Officer Course
- FA Advance Officer Course
- FA Basic Officer Leadership Course 3
- Basic Officer Leadership Course 2
- Officer Candidate Course

CIVILIAN SCHOOLING:

- Master of Arts in Mathematics Education
- Bachelor of Secondary Education in Mathematics

ICE BREAKER

Jumbled Letters	Correct Answer
WZTHIOER	HOWITZER
SCOEL RIA PSOPTRU	CLOSE AIR SUPPORT
LAVAN NUG RIEF	NAVAL GUN FIRE
YTRILLEART	ARTILLERY
RIEF PUPROST	FIRE SUPPORT
ORTRMA	MORTAR

NEED

The subject provides Signal Officers and NCOs with a fundamental understanding of artillery operations to enable them to effectively plan, coordinate, and provide communication support to Field Artillery units and integrate communication systems into fire support operations.

CLASSROOM RULES













SAFETY RULES







DURATION



REFERENCES

- *SP 3-043*
- *Field Artillery Technical Handbook, HPA dtd August 2005*

ROADMAP

EXPLAIN THE FOLLOWING;

- *FA Operations*
- *Missions & System functions*
- *Tactical Roles and desired effects*
- *Capabilities and Limitations*
- *Fires Support Environment*
- *Operations Fundamental*

Summarize;

Assess; WE

TERMINAL LEARNING OBJECTIVE

At the end of this subject, the students shall be able to apply basic knowledge of Field Artillery operations to determine appropriate communication support requirements for effective command and control and coordination of fire support operations.*(C3)*

ENABLING LEARNING OBJECTIVES

- *Discuss the FA Operations, missions, and system functions. (C3)*
- *Discuss the functions tactical roles and desired effects. (C3)*
- *Discuss the capabilities and limitations (C3)*
- *Discuss the fires support environment(C3)*

CLASSROOM RULES









SAFETY PRECAUTIONS







FIELD ARTILLERY OPERATION

MISSION

To destroy, neutralize or suppress the enemy with timely and accurate field artillery fires in support of ground forces

and to assist in the integration of all fire support means into combined arms operation

Functions and Tactical Roles

Functions

- *Provide continuous, accurate, and timely field artillery fire support to ground forces.*
- *Provide its component communication, mobility requirements, survey, meteorological data, target acquisition, and fire direction system.*
- *Establish the nucleus of fire support cells of the supported maneuver forces of the Army and provide fire support elements to its maneuver units.*
- *Command and control additional field artillery units.*

Tactical Roles

Close support *– close support fires are fires used to engage enemy troops, weapons, or positions that are threatening or can threaten the force in either the attack or the defense.*

Attack at depth *(Interdiction) – artillery fires that attack at depth are used to disrupt, delay, and destroy enemy forces that, because of range limitation or intervening terrain, cannot fire their primary weapon system on friendly forces. Attacking at depth creates "windows" for friendly unit's offensive maneuver.*

Counter fires *– consist of fires targeted throughout the battlefield that are intended to attack the total enemy fire support system.*

Capabilities and Limitations

Capabilities

- *FA can provide fires under all conditions of weather and in all types of terrain.*
- *FA can shift and mass fires rapidly without requirement to displace.*
- *FA can add depth to the battlefield.*
- *FA can fire variety of shell and/or fuze combinations.*
- *FA can provide continuous support by judicious displacement.*

Limitations:

- *FA has limited self-defense capability against ground or air attack.*
- *FA has limited capability to destroy point targets without considerable ammunition expenditure.*
- *FA has a firing signature that makes it vulnerable to detection by enemy target acquisition assets.*
- *FA weapons can only effectively engage targets within their specified range limits.*

CLASSIFICATION OF FA WEAPONS

AS TO CALIBER

- *LIGHT - 120mm & LESS*
- *MEDIUM - 121mm THRU 160mm*
- *HEAVY - 161mm THRU 210mm*
- *VHEAVY - 211mm & MORE*

CLASSIFICATION OF FA WEAPONS

AS TO MEANS OF TRANSPORT

- *Towed*
- *Self Propelled*
- *Aerial*

AS TO MEANS OF TRANSPORT

- *Towed*



AS TO MEANS OF TRANSPORT

- SELF PROPELLED



AS TO MEANS OF TRANSPORT

- *AERIAL*



Effects of Fires

***Destroy** - a tactical mission task that physically renders an enemy force combat-ineffective until it is reconstituted.*

In the context of the computed effects of field artillery fires, destruction renders a target out of action permanently, or ineffective for a long period of time, producing at least 30-percent casualties or materiel damage.

Effects of Fires

Neutralize - *a tactical mission task that results in rendering enemy personnel or materiel incapable of interfering with a particular operation.*

Neutralization in the context of the computed effects of field artillery fires renders a target ineffective for a short period of time, producing at least 10-percent casualties or materiel damage.

Effects of Fires

Suppress - a tactical mission task that results in temporary degradation of the performance of a force or weapons system below the level needed to accomplish the mission.

Suppression, in the context of the computed effects of field artillery fires, renders a target ineffective for a short period of time producing at least 3-percent casualties or material damage.

Effects of Fires

Disrupt *- a tactical mission task in which a commander integrates direct and indirect fires, terrain, and obstacles to upset an enemy's formation or tempo, interrupt the enemy's timetable, or cause enemy forces to commit prematurely or attack in a piecemeal fashion.*

Defeat *- a tactical mission task that occurs when an enemy force has temporarily or permanently lost the physical means or the will to fight.*

Diversion *- is the act of drawing the attention and forces of an enemy from the point of the principal operation; an attack, alarm, or feint that diverts attention.*

Description Of Artillery Fires Operational Environment

The Philippine Archipelago, which has distinct land features, dictates that its artillery assets must be capable of displacing rapidly from one area or island to another without disrupting the multidimensional operations.



OPERATIONS FUNDAMENTALS

Basic tasks

Before FA uses seven basic tasks but it is now summarized into four (4) basic tasks only to support maneuver forces on the battlefield. The following are basic tasks that the FA must do:

a. Support Forces in Contact. *FA fires must be responsive and ready to provide when requested by the supported unit*

b. Support the Concept of Operations. *FA fires must fulfill the objective of the supported maneuver unit commander.*

c. Synchronize Fire Support. *FA fires needs to be synchronized with other available fire support assets to economize resources with maximize effect. d. Sustain FA Operations. FA commanders at all levels must sustain FA units.*

System

The FA system is composed of different functional elements that are necessary to produce the needed effects in combat to support the maneuver commanders' concept of operation. These functional elements are:

a. Target Acquisition
b. Fire direction and coordination
c. Weapon/ammunition combinations
d. Organization
e. Survey
f. Ballistic Meteorology
g. Communications
h. Logistics
i. Mobility
j. Employment Tactics

Target Acquisition

Target acquisition is the "eyes and ears" of the system. It is responsible for the effective employment of fires through timely and accurate detection, identification, and location of targets. The Target Acquisition Platoon is under the Headquarters and Headquarters Battery in the FABn Table of Organization and Equipment. It forms part of the FA target intelligence that collects, process and disseminates information. Most common collectors of information for FA intelligence are artillery observers, through ground, air, sound, and flash, and vertical air photography like drones. In large operations the Intelligence Office of the FA Regiment and FA Bn are the principal unit in artillery intelligence.

Fire direction and coordination

Fire direction center and coordination is the "brain" of the system that processes appropriate FA fires needed to attack targets quickly and effectively. Fire direction center is responsible to convert call for fire of the forward observer into appropriate fire commands. Fire coordination is the planning and execution of fire to adequately cover targets with a suitable weapon or group of weapons.

Weapon/Ammunition Combinations

Weapon/ammunition combinations is the "muscle" of the system. The right combination of weapon and ammunition makes the fire effective and efficient

Organization

The establishment of command and/or support relationship for cannon, rocket, and missile systems organization with maneuver unit gives accurate fusion of indirect fire capacities for the effective and economical delivery of FA fires. The organization of the FA units are shown in detail in Tables of Organization and Equipment (Table of Organization and Equipment (TOE), 2019). An FA Platoon is the smallest artillery unit that will be deployed to provide fire support to maneuver units.

Survey

FA survey operations provide a common grid of location and direction for accurate delivery and massing of fires. Survey section is an organic unit of FABn under Headquarters and Headquarters Battery.

Ballistic Meteorology

Ballistic meteorological data are necessary for accurate FA fires. Meteorological (MET) information is one of the five requirements for accurate predicted fires. MET section is an organic unit of FABn under Headquarters and Headquarters Battery.

Communications

The effectiveness and responsiveness of FA fires greatly relies on the efficiency and reliability of communication. The principal communication of FA unit is radio for voice and digital. FA units are issued with Harris radios except for 155mm SP unit which is equipped with manet radio. The Signal platoon is an organic unit of FABn under Headquarters and Headquarters Battery.

Logistics

Logistics support is designed to sustain the conduct of FA operations

Mobility

FA units are issued with mobility assets like KM450, KM250, KM500, M35 and general-purpose vehicles for unit mission accomplishment.

Employment Tactics

The mission of the supported unit dictates the employment of the supporting FA. During offense, the supporting FA unit is positioned well forward to support the attack and exploit the range of its weapons. While in the defense, supporting FA unit is positioned at the rear to ensure continuity of FA fires in the event of enemy penetration.

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CHECK ON LEARNING

SUMMARY

- *FA Operations*
- *Missions & System functions*
- *Tactical Roles and desired effects*
- *Capabilities and Limitations*
- *Fires Support Environment*
- *Operations Fundamentals*

ASSESSMENT

WRITTEN EXAMINATION


QUESTIONS?



End of Presentation